

A Resilient and Power-Efficient Automatic Power-Down Sense Amplifier for SRAM Design

Ozan Ekame Pekgoz and William Borland
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Instructor: Prof. Rabia Yazicigil Kirby

I. INTRODUCTION

Sense amplifiers in SRAM detect the small voltage difference between the bitlines (BL and BLBAR) during a read, and amplify it to a full logic level. There are a few ways to design them, and each has its own trade-offs and drawbacks. The latch-type sense amplifier is fast and consumes little power, but it can fail when the bitline swing is too small. If the voltage difference isn't large enough by the time the sense enable signal fires, the latch can lock onto the wrong value. This issue would be particularly relevant at the fast fast (FF) corner, where the device would be operating at a lower temperature, thus driving more current with the same gate voltage, essentially operating at a lower V_t . With these conditions, a mismatch ΔV_t between the devices is more likely to cause a current imbalance sufficient to flip the latch erroneously. The current-mirror design does not face the same sensing failure as the latch-type sense amplifier, but the downside is that it draws current continuously while on, wasting power. The goal of this project was to recreate a current-mirror sense amplifier with an automatic power-down (APD) circuit that shuts the sense amplifier off as soon as sensing is complete, following the designs from our paper. This provides the reliability of the current mirror without drawing unnecessary power. Furthermore, we replicate the paper's design of a dual V_{HL} APD module to avoid APD failure caused by process variations in Schmitt trigger thresholds required for APD functionality. This would allow the designer to set two different V_{HL} thresholds (one low, one high) on two different Schmitt trigger modules, selecting between the two after fabrication with a signal to ensure that one is within the correct operating range.

II. BACKGROUND

To understand the limitations of each approach, it helps to first look at how an SRAM read operation works. The memory cell used in this design is a standard 6T SRAM cell with precharge transistors. At the start of each read cycle, both bitlines are pulled up to half V_{DD} . When the wordline goes high, the access transistors turn on, and the bitlines charge or discharge based on what's stored in the cell. This creates a small voltage difference between BL and BLBAR, which is fed into a sense amplifier to amplify it to full V_{DD} or ground.

The latch-type sense amplifier [Figure 1] uses cross-coupled inverters to amplify a small voltage difference through positive feedback. It works correctly when the bitline swing is large, but when the swing is small, it can latch the wrong value before the true differential builds up. The minimum sensing margin can be found either through binary search in simulation, or analytically using Pelgrom's mismatch model, where

$$\sigma(\Delta V_t) = \frac{A_{vt}}{\sqrt{W \times L}} \quad (1)$$

relates transistor sizing directly to input offset voltage. In our simulation [Figure 1], this threshold was approximately 18 mV, corresponding to a rise time of $T_R \approx 1.5$ ns. The current-mirror sense amplifier [Figure 3] avoids this problem by nature, as it does not have the positive feedback loop the latch does. The design also includes an NMOS source follower level shifter, which lowers the bitline voltage, giving the PMOS transistors in the main amplifier sufficient headroom to operate when the bitlines are precharged to full V_{DD} . For our implementation, we included it to match the paper's proposed design. It functions by passing the level-shifted SO and SOBAR to the gates of two NMOS devices, each in the main amplifier. Along with four more PMOS devices, these form a differential amplifier, one seeing the difference SO-SOBAR, the other seeing SOBAR-SO. This provides the current necessary to pull SAOUT and SAOUTBAR to their desired levels. The issue with the current mirror amplifier design is that the circuit draws a steady current while SE is high, resulting in significant power consumption. Even after the output has settled, the sense amplifier continues to draw current for the remainder of the read cycle, which is the inefficiency that the APD design is meant to address.

III. PROPOSED APD SENSE AMPLIFIER

Our Automatic Power Down block design [Figure 4] uses the current-mirror sense amplifier as the core, but adds an automatic power-down circuit [Figure 6] that cuts the current once the power-down signal goes low (active low), activating PMOS devices, which pull both SAOUT and SAOUTBAR high, preventing current flow. The APD block consists of three parts: a pair of Schmitt-trigger buffers, a high-skew NAND gate, and a dynamic inverter.

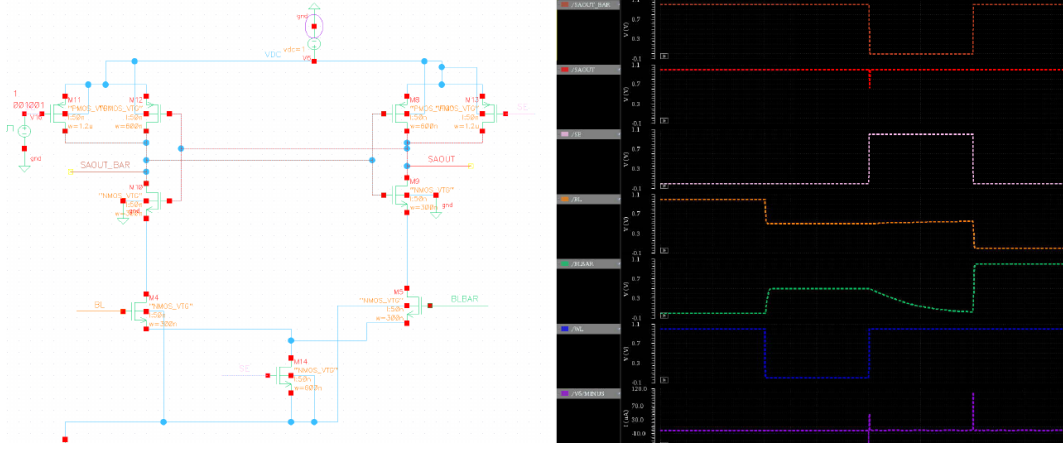


Fig. 1: Our implementation of the Latch type sense amplifier and results from a waveform.

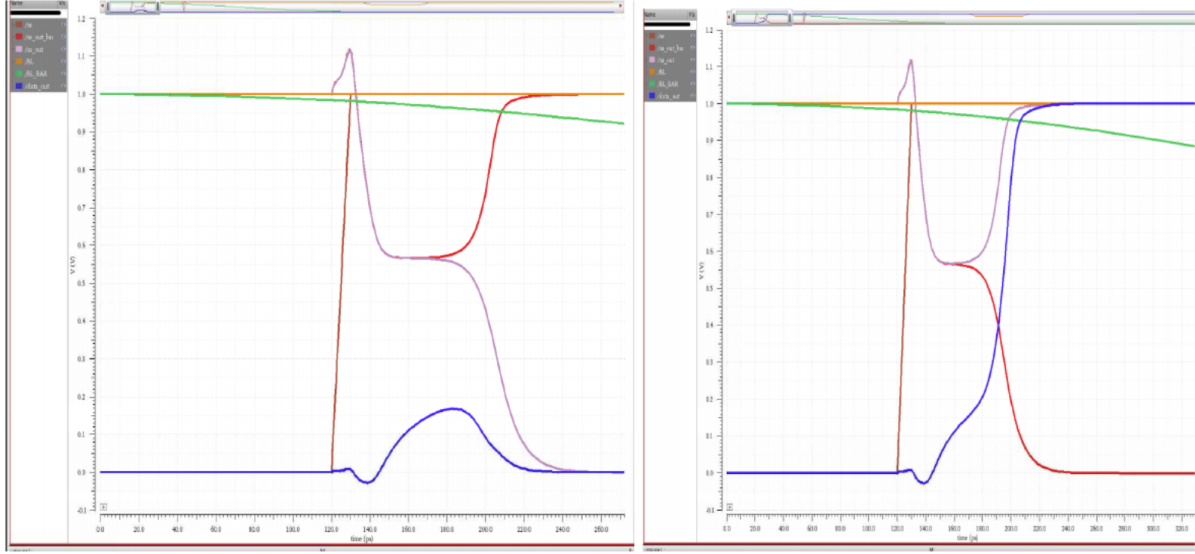


Fig. 2: results from our test module showing a sensing failure (left) compared to a successful sense (right) in a latch-type amplifier.

The Schmitt trigger buffers [Figure 5] are the heart of the power-down block, as they sense when SAOUT or SAOUT-BAR crosses V_{HL} , the Schmitt trigger's rising threshold. By nature, the Schmitt trigger we implemented was inverting, so our implementation added an inverter between SAOUT/SAOUTBAR and the Schmitt trigger to resolve the issue. Once the Schmitt trigger's V_{HL} threshold is reached, its output will switch (high to low), and the inputs seen by the NAND2 will be (1,0) rather than (1,1), which raises its output, finally reaching a dynamic inverter that will see high at its NMOS gate input, pulling out a power-down signal low, powering off our main amplifier.

IV. DESIGN PARAMETERS AND SIZING

We used a V_{DD} of 1 V and a length of 50 nm for all devices. For our 6T SRAM cell, we used standard sizings. 180 nm W_p , 90 nm W_n for the inverters, and 120 nm W_n for

the access transistors. We did not observe any read upsets with these parameters because our access transistors were not strong enough to overwrite the cell on a read. We used large ($W_p = 10\mu$, $W_n = 5\mu$) precharge and discharge devices to ensure fast reads and writes, so we would not bottleneck our design with the SRAM cell. The most important sizing constraints for our sense amplifiers came from the paper, which stated that our amplifier designs should meet certain sizing requirements. For example, the latch type should have a total silicon cost of 40 units, and the APD current mirror should have a total silicon cost of 51 units. In our case, one unit was 90 nm because that was our minimum sizing for our SRAM NMOS devices. In sizing each individual transistor in the latch-type (Figure 1), we began by assigning weights W to the devices based on how strong we thought they would need to be for operation, assuming our standard PMOS/NMOS ratio would be 2W:1W. We kept the cross-coupled inverters at 2W/1W,

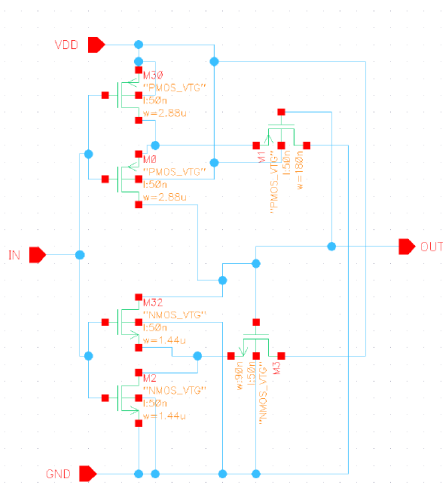


Fig. 5: Our implementation of a CMOS Schmitt trigger from reference [2].

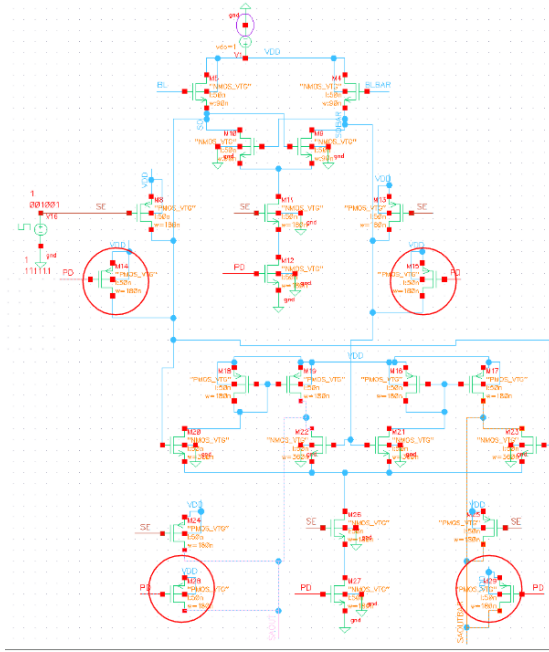


Fig. 6: Our implementation of the APD Current mirror sense amplifier

We made M3 (and the corresponding PMOS devices) unit width, and scaled M2 (and the corresponding PMOS device) accordingly, giving us the sizings seen in Figure 5. These are the most important devices to tune when tuning this Schmitt trigger's V_{HL} threshold. When designing the tri-state Schmitt trigger buffers (high and low) [Figure 7], we used the design from the paper directly, with unit-width/length devices.

V. PROCESS VARIATION AND DUAL-THRESHOLD DESIGN

One major issue with the APD circuitry we have described is that the Schmitt triggers can fail under significant process

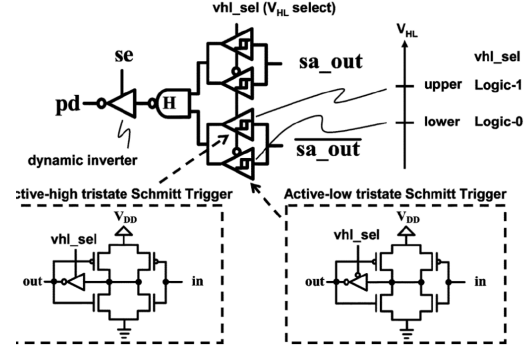


Fig. 7: Dual V_{HL} select Power Down Architecture.

TABLE I: Power and delay comparison.

Configuration	Power	Delay
Current-Mirror SA (no APD)	123.6 μ W	3.1 ns
Current-Mirror SA + APD	22.9 μ W	3.1 ns

variations. Specifically, as the paper notes, if the variation in the Schmitt trigger V_{HL} exceeds $\pm 10\%$, one Schmitt trigger pair would malfunction. The solution is a dual- V_{HL} APD circuit, as described in the paper, which uses two parallel Schmitt trigger pairs with higher and lower V_{HL} thresholds, respectively. As we briefly introduced, after fabrication, the designer can choose the higher- or lower- V_{HL} Schmitt trigger pair based on the chip's corner. For example, if V_{HL} rises due to process variations, the lower- V_{HL} pair is selected with the select signal.

VI. RESULTS

We were able to replicate the waveforms from the paper in Figure 2, depicting an incorrect latch-type amplifier read and a successful read, respectively. In these graphs is the SAOUT, which should be pulled high for a read of 1. We simulated all three configurations - latch-type, current-mirror without APD, and current-mirror with APD - and compared the power and delay of the current mirrors with and without APD, as those were of most interest to us. These Cadence waveforms are seen in Figure 1, Figure 3, and Figure 8. The results are shown in Table I. Qualitatively, these waveforms all show the bitlines changing slightly, propagating their difference to the SAOUT/SAOUTBAR, and in the case of the APD version of the current mirror, the power down signal being pulled down, and the current from V_{DD} being cutoff.

Adding the APD circuit reduced power from 123.6 μ W to 22.9 μ W, about an 81% reduction, while keeping the same 3.1 ns delay. The paper reports, over a range of frequencies, 28%-89% power reduction by adding the APD block to the current mirror - a range that includes our data. The delay didn't change because the PD signal fires only after the output has settled, so it doesn't affect the timing path. This again matches the paper's results, which showed a 5% increase in delay when adding the power down block, which is an insignificant increase.

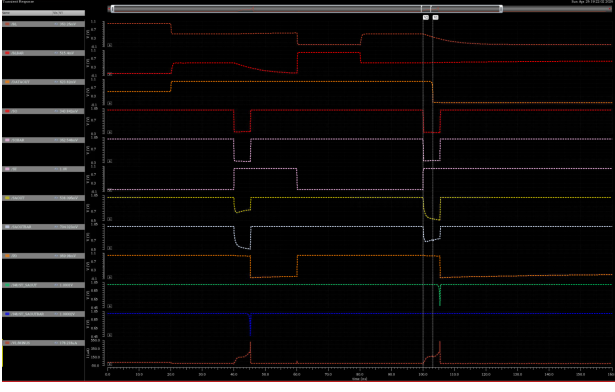


Fig. 8: Waveform results using APD sense amplifier.



Fig. 9: Waveform for our implementation of dual V_{HL} APD.

A direct comparison of the current drawn between our two waveforms as shown in Figure 10, which shows the immediate cutoff from the APD compared to the consistent current drawn without it. The latch-type sense amp uses much less power; we calculated it at about 217 nW. Again, it's not reliable for small bitline swings, so it does not make sense to compare the two styles directly. We also verified the dual-threshold circuit with DC sweeps of the V_{HL} selection and transient simulations of the full APD block, and it behaved correctly across read cycles. This can be seen in Figure 9, which depicts the waveform output of the Dual V_{HL} , actually connected to our APD current mirror SA. This waveform shows behavior similar to the normal APD network, with slightly different timing for the PD signal pull-down, which we can attribute to the different styles/sizings of the Schmitt triggers.

VII. CONCLUSION

Our implementation of both the standard and Dual V_{HL} successfully powered off our current mirror sense amplifier, and the former reduced power consumption by 81% without increasing delay. This amplifier design solves the problem of a standard current mirror by cutting power consumption after the read has completed, and avoids sensing failure, which can affect latch-type amplifiers. Our results match the findings of

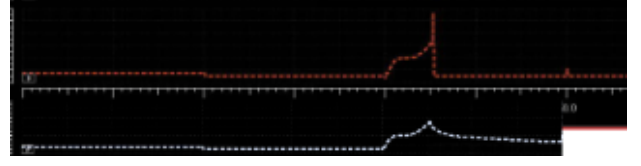


Fig. 10: direct comparison between the shape of the current waveform for APD (orange) vs no-APD (white) current mirror sense amplifiers.

the paper for all the devices we recreated, with differences that can be mainly attributed to our sizing choices and the different technology nodes we used compared to the paper.

REFERENCES

- [1] Y.-C. Lai and S.-Y. Huang, "A Resilient and Power-Efficient Automatic-Power-Down Sense Amplifier for SRAM Design," *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 55, no. 10, pp. 1031–1035, Oct. 2008.
- [2] I. M. Filanovsky and H. Bakes, "CMOS Schmitt Trigger Design," *IEEE Transactions on Circuits and Systems I: Fundamental Theory and Applications*, vol. 41, no. 1, pp. 46–49, Jan. 1994.